

Canvass Arithmetic: From Unofficial Returns To Certified Totals

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Abstract

Election totals change for legitimate reasons between election night and certification: late mail ballots arrive, provisional ballots are adjudicated, write-ins are reviewed, duplicates are resolved, and recounts or court orders may alter the public record. This paper describes how RCOUNT represents that evolution without treating every change as either fraud or final truth. We define canvass arithmetic as the collection of status-specific summary checks, jurisdiction rollups, public status events, batch accounting equations, and source hash checks that let third parties replay public count claims. Using the `canvass-correction`, `mail-batch-added`, and `missing-batch` synthetic fixtures, we show how a verifier can identify the layer where a canvass claim passes or fails while leaving legal certification to election officials and courts.

1 Introduction

Election-night totals are not certification. They are preliminary public claims, often produced before every ballot category has been accepted, duplicated, adjudicated, or reconciled. A trustworthy public count system must therefore explain change, not merely detect change. The same one-vote delta can be a write-in adjudication correction, a late mail batch, a data-entry error, a missing source file, or an unexplained alteration. These cases require different evidence and different public conclusions.

V.0 introduced RCOUNT as a reproducible package format for election-count claims. This paper focuses on the canvass arithmetic layer: how unofficial, canvassed, recounted, amended, and certified snapshots can be represented as status-specific summaries, how public events explain transitions, and how batch manifests and jurisdiction totals constrain the arithmetic.

The key boundary remains the same. RCOUNT does not decide that a canvassing board acted lawfully. It can show that a declared canvassed total equals the published precinct or batch summaries; that a correction event exists; that a late mail batch reconciles accepted, counted, and rejected ballots; and that the source files still match their hashes. That is a mechanical replay of public evidence, not legal certification.

2 Status-Specific Snapshots

RCOUNT treats status as part of the summary key. A contest summary is not just “the total for P-001”; it is the total for a contest, reporting unit, optional batch, and status. This lets an unofficial total and a canvassed total coexist in the same package without overwriting one another.

Status	Example source	Mechanical checks
Unofficial	Election-night export	Contest arithmetic, jurisdiction rollup for unofficial summaries, source hash.
Canvassed	Canvass report or corrected export	Contest arithmetic, jurisdiction rollup for canvassed summaries, status-event evidence.
Recounted	Recount report	Same arithmetic on the recount snapshot plus recount event lineage.
Amended	Amended statement or court order	Arithmetic on amended snapshot plus public event reference.
Certified	Certification artifact	Consistency of records marked certified, not legal validity.

The jurisdiction-total check is status-specific. If a package contains both unofficial and canvassed snapshots, the unofficial jurisdiction total must equal the unofficial precinct summaries, and the canvassed jurisdiction total must equal the canvassed precinct summaries. This prevents a verifier from comparing records across statuses as if they belonged to one immutable table.

3 Correction Events

The `canvass-correction` fixture models a small but important case. The unofficial snapshot reports one fewer vote for Candidate B in precinct P-001 than the canvassed snapshot. The package includes both sets of summaries and a public correction event from unofficial to canvassed status.

Snapshot	A	B	WI	Under	Over	Blank	Counted
P-001 unofficial	40	34	1	3	1	0	79
P-001 canvassed	40	35	1	3	1	0	80
Jurisdiction unofficial	65	64	1	7	1	1	139
Jurisdiction canvassed	65	65	1	7	1	1	140

The local equation for P-001 therefore changes by one declared vote, not by a hidden overwrite:

$$40 + 34 + 1 + 3 + 1 + 0 = 79 \quad \rightarrow \quad 40 + 35 + 1 + 3 + 1 + 0 = 80.$$

The verifier checks three things. First, each summary satisfies local contest arithmetic: candidate votes plus residual counts equal counted ballots. Second, each status-specific

jurisdiction total equals the sum of its reporting units. Third, a correction event exists with authority, source references, explanation, and a transition from unofficial to canvassed.

The minimal normalized event contract is deliberately small:

Field	Meaning
<code>event_id</code>	Stable identifier within the package.
<code>event_type</code>	Typed reason, such as correction, late-mail-batch-added, recount-update, court-order, or certification.
<code>status_before</code> , <code>status_after</code>	The declared transition. Initial unofficial reports may have the same before and after status.
<code>effective_at</code>	Public event date or timestamp asserted by the source adapter.
<code>authority</code>	Office, board, court, or source authority making the public assertion.
<code>source_refs</code>	Package source identifiers whose raw hashes bind the event to declared evidence.
<code>explanation</code>	Human-readable reason for the transition.

A compact record for this fixture is:

```
{
  "event_id": "event-0002",
  "event_type": "correction",
  "status_before": "unofficial",
  "status_after": "canvassed",
  "effective_at": "2024-11-12T18:22:00Z",
  "authority": "SYN County Canvassing Board",
  "source_refs": ["source:canvass-minutes-2024-11-12"],
  "explanation": "Canvass correction added one Candidate B vote in P-001..."
}
```

This does not prove that the correction was lawful. It proves that the package does not hide the delta. A public reader can see the before snapshot, the after snapshot, the event metadata, and the source hashes that bind the package to its declared evidence. A passing RCOUNT event check means "this transition is declared and linked," not "this board action satisfies every statute."

4 Batch Accounting

Canvass changes often enter through batch accounting rather than direct precinct edits. Late mail ballots, provisional adjudication, duplicated damaged ballots, and central-count batches may be accepted or rejected after election night. RCOUNT models this with batch manifests and optional batch-specific summaries.

For every batch manifest, accepted ballots must reconcile:

$$\text{accepted ballots} = \text{counted ballots} + \text{rejected ballots}.$$

These are declared source facts. RCOUNT verifies that the declared counts are internally consistent; it does not decide whether a voter was eligible, whether a cure was timely, or

whether a provisional ballot should have been accepted. For every batch summary, the summary’s counted ballots must match the manifest’s counted ballots:

$$\text{summary counted ballots} = \text{manifest counted ballots}.$$

Batch-specific summaries are adapter-dependent: a package may contain ordinary reporting-unit summaries without batch ids, but once a summary references a batch id, the corresponding batch manifest is required.

In the **mail-batch-added** fixture, P-001 has an election-day batch and a late mail batch. The late mail batch is not suspicious because it is late; it is auditable because it is declared. The event says a late-arriving mail batch was accepted before canvass, the batch manifest declares accepted/counted/rejected ballots, and the batch summary ties contest totals to that accounting evidence.

Batch	Accepted	Counted	Rejected	Check
P-001 election day	70	70	0	Accepted equals counted plus rejected.
P-001 late mail	10	10	0	Batch summary must count 10 ballots.
P-002 election day	60	60	0	Batch summary must count 60 ballots.

The negative **missing-batch** fixture removes the late mail manifest while leaving a summary that references it. Arithmetic alone may still look plausible, but the accounting evidence is missing. The verifier should therefore fail at **batch_summary_total**, not at a generic package error.

This paper stops at summary and batch accounting. CVR-to-summary reconciliation, including ballot cards, CVR rows, and parser diagnostics, belongs to V.6.

5 Failure Attribution

Canvass arithmetic is useful only if failures are attributed to the right layer. RCOUNT transcripts therefore name equation ids rather than producing only a single pass/fail bit.

Layer	Failure	Interpretation
Contest sum	Candidate votes plus residuals do not equal counted ballots	Local summary arithmetic is wrong.
Jurisdiction total	Jurisdiction total does not equal reporting-unit summaries	Rollup is wrong for that contest/status.
Canvass correction	Changed canvass lacks public correction event	Transition evidence is missing.
Batch summary	Batch summary references absent manifest	Accounting evidence is missing.
Accepted ballots	Accepted ballots differ from counted plus rejected	Batch manifest accounting is inconsistent.
Source hash	Raw source file changes after packaging	Source evidence no longer matches the package index.

In verifier transcripts these layers correspond to stable machine ids:

```
contest_selection_sum
jurisdiction_contest_total
canvass_correction_event
batch_summary_total
accepted_ballots
source_hash_match
```

This attribution matters rhetorically as well as technically. A source hash failure is not the same as a bad jurisdiction total. A missing batch manifest is not the same as a fraudulent canvass. RCOUNT's contribution is to make those distinctions visible enough that later legal, administrative, or audit decisions can reason from a stable mechanical record.

6 Verifier Traceability

The examples above are L1 package fixtures. They are regenerated by `rcount-io` examples, verified by `rcount-cli`, and summarized by `rcount-audit` transcripts.

```
cargo run -p rcount-io --example canvass_correction_package
cargo run -p rcount-io --example mail_batch_added_package
cargo run -p rcount-io --example missing_batch_package

cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/canvass-correction
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/mail-batch-added
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/missing-batch
```

Fixture	Expected result	Checks that carry the paper claim
canvass-correction	pass	contest_selection_sum, jurisdiction_contest_total, canvass_correction_event, source_hash_match.
mail-batch-added	pass	accepted_ballots, batch_summary_total, jurisdiction_contest_total, source_hash_match.
missing-batch	fail	batch_summary_total fails be- cause a summary references batch:P-001:late-mail with no manifest.

A transcript excerpt for the negative fixture is intentionally specific:

```
{
  "status": "fail",
```

```

    "checks": [
      {
        "equation_id": "batch_summary_total",
        "status": "fail",
        "error": "summary ... references missing batch id: batch:P-001:late-mail"
      },
      {"equation_id": "jurisdiction_contest_total", "status": "pass"},
      {"equation_id": "source_hash_match", "status": "pass"}
    ]
  }

```

That pattern is the point of canvass arithmetic. A package can preserve source hashes and jurisdiction totals while still failing because one batch-level piece of evidence is absent. Public readers get the failure layer, not a vague accusation.

7 Boundaries

RCOUNT canvass arithmetic has four main boundaries.

First, it does not decide ballot eligibility. A package can record accepted, counted, rejected, and cured ballot categories, but the legal decision to accept or reject a ballot is made under jurisdiction-specific law.

Second, it does not replace canvassing-board minutes or certification records. Status events should point to public authority and source references, but the software verifies the presence and consistency of those records, not their legal sufficiency.

Third, it does not require one national canvass workflow. The synthetic fixtures are deliberately small and jurisdiction-neutral. Real adapters must preserve state and local differences in cure periods, provisional rules, recount thresholds, and reporting-unit conventions.

Fourth, it does not expose voter choices. Even when a future adapter ingests CVR or ballot-manifest evidence, public RCOUNT packages must preserve the privacy boundary described in V.0. Public event metadata also inherits that boundary: small reporting units, rare write-ins, ballot-style combinations, timestamps, or overly precise source references can create reidentification risk. Hashing a source file provides tamper evidence; it does not make private source data safe to publish.

Finally, V.1 is not an end-to-end cryptographic voting protocol. It supplies replayable public arithmetic and evidence references. Later papers may add privacy-safe inclusion proofs and CVR reconciliation, but the rule remains: prove inclusion, never prove candidate choices.

8 Conclusion

Canvass arithmetic is the discipline of replaying public count evolution without collapsing it into a single final table. RCOUNT represents status-specific snapshots, jurisdiction rollups, status events, batch manifests, and source hashes so that changes can be explained and failures can be attributed to the right layer. The `canvass-correction`, `mail-batch-added`, and `missing-batch` fixtures are small, but they establish the core contract: changed totals

need public lineage, batches need accounting evidence, and mechanical verification must stop short of legal certification.

The same boundary protects voters. RCOUNT can make a public count package auditable, but it must not turn canvass evidence, hashes, or future inclusion proofs into a way to prove how a person voted.

References

Gio Della-Libera. Rcount: Reproducible election count packages, 2026a. V.0 working paper in `research/tracks/V-election-audit`.

Gio Della-Libera. Rcount substrate specification, 2026b. Working specification in `docs/specs/2026-05-12-rcount-substrate.md`.